



Aror University of Art, Architecture, Design & Heritage Sukkur.

Department of Artificial Intelligence and Multimedia Gaming

Programming for AI (Spring-2026)

LAB No. 09

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Objective of Lab No. 09:

After performing lab 9, students will be able to:

- 1. Use Pandas to perform Data Cleaning**
- 2. Use Pandas to perform Data Filtering**
- 3. Use Pandas to perform Data Analysis**

Lab Tasks:

Task 01: Load Titanic Dataset, How many rows and columns are there in this dataset?

Task 02: What are the data types of each column? And Which columns contain missing values?

Task 03: Find the average age of passengers, Find the maximum fare, count how many passengers are recorded (Don't use Query Method or Data Filtering)'

Task 04: Extract and Display Pclass, Age, and Fare Columns

Task 05: Display first 10 rows with columns Name, Sex, Age using loc

Task 06: Data Analysis:

- 1. Find passengers older than 25**
- 2. Find female passengers who survived**
- 3. Find passengers in 1st class OR fare greater than 50**



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4. Find male passengers under age 20

Task 07: Display:

- o Count number of males and females
- o Count passengers in each class
- o Count survived vs not survived

Task 08: Data Cleaning:

- ☐ Identify columns with missing values
- ☐ Fill Age with mean
- ☐ Fill Embarked with mode
- ☐ Apply forward fill on Cabin